A telescope made of squares

$L_k(x^2 + x) = x^{2^k} + x$, read slowly

Abstract

One short induction.

- `truncTrace_artin_schreier`: $L_k(x^2 + x) = x^{2^k} + x$.

Here $L_k(u) = \sum_{i<k} u^{2^i}$ is the *truncated trace*. The identity is the bridge that rewrites the Kasami differential as a truncated trace — the entry point to the Dempwolff–Müller Theorem 3.2 engine.

Setting

$$\Gamma = \big(F : \text{Type},\ [\text{CommRing } F],\ [\text{CharP } F\ 2]\big)$$

No finiteness needed — only $1 + 1 = 0$. The truncated trace is

$$L_k(u) \;=\; \sum_{i=0}^{k-1} u^{2^i} \;=\; u + u^2 + u^4 + \cdots + u^{2^{k-1}}.$$

1 The lemma

$$\forall k : \mathbb{N},\ \forall x : F. \qquad L_k(x^2 + x) \;=\; x^{2^k} + x$$

```
lemma truncTrace_artin_schreier {F : Type*} [CommRing F] [CharP F 2]
    (k : ℕ) (x : F) :
    truncTrace k (x ^ 2 + x) = x ^ (2 ^ k) + x := by
  induction' k with k ih generalizing x <;>
    simp_all +decide [truncTrace, pow_succ, pow_mul]
  · rw ←[ two_smul F x, CharTwo.two_eq_zero, zero_smul]
  · rw [Finset.sum_range_succ, ih]
    rw [add_pow_char_pow, mul_pow]; ring
```

```
  simp +decide [show (2 : F) = 0 by exact CharP.cast_eq_zero F 2]
```

Why it telescopes

Two facts in char 2:

- **Additivity** (Freshman's Dream): L_k is additive, since each $u \mapsto u^{2^i}$ is. So $L_k(x^2+x) = L_k(x^2) + L_k(x)$.
- **Square commutes with L_k**: $L_k(x^2) = L_k(x)^2$, because squaring is the Frobenius ring hom.

Write $T = L_k(x)$. Then

$$L_k(x^2 + x) = T^2 + T.$$

But $T^2 + T$ is itself a telescoping sum:

$$T^2 = \sum_{i=0}^{k-1} x^{2^{i+1}} = \sum_{i=1}^{k} x^{2^i}, \qquad T = \sum_{i=0}^{k-1} x^{2^i},$$

$$T^2 + T = \Big(\sum_{i=1}^{k} x^{2^i}\Big) + \Big(\sum_{i=0}^{k-1} x^{2^i}\Big) = x^{2^k} + x,$$

because every middle term x^{2^i} $(1 \le i \le k-1)$ appears *twice* and $u + u = 0$. Only the endpoints x^{2^k} and $x^{2^0} = x$ survive.

$$\begin{array}{lcccccc}
T^2: & & x^2 & x^4 & \cdots & x^{2^{k-1}} & x^{2^k} \\
 & & | & | & & | & \\
T: & x & x^2 & x^4 & \cdots & x^{2^{k-1}} &
\end{array}$$

Matched pairs cancel (char 2); only x^{2^k} (top right) and x (bottom left) remain.

This is the additive shadow of a geometric series. Over $\mathbb{R}$, $\sum_{i<k} r^i = (r^k - 1)/(r-1)$; here $r =$ "square", the denominator $r - 1$ is "square plus identity" $= u \mapsto u^2 + u$, and the identity says $L_k \circ (u^2 + u)$ recovers $u^{2^k} + u$, the numerator.

2 Where it lands: the Kasami differential as a trace

The Kasami collision analysis needs to recognise $x^{2^k} + x$ as L_k(something). The lemma says that "something" is the Artin–Schreier input $x^2 + x$:

$$x^{2^k} + x \;=\; L_k(x^2 + x).$$

This is exactly the rewrite used (twice) to turn a difference of d-th powers into a truncated trace, feeding the Dempwolff–Müller decomposition $\Phi(u) = L_k(u)^{q+1}/u^q$.

```
rw [truncTrace_artin_schreier, truncTrace_artin_schreier]
```

One-screen summary

step	**content**
additive	$L_k(x^2 + x) = L_k(x^2) + L_k(x)$
Frobenius	$L_k(x^2) = L_k(x)^2$
telescope	$T^2 + T = x^{2^k} + x$ (middle terms cancel, char 2)
land	rewrite Kasami difference $\to$ truncated trace

$L_k(x^2 + x) = x^{2^k} + x \;\Rightarrow\;$ Kasami differential = truncated trace $\;\Rightarrow\;$ Dempwolff–Müller engine.